

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919588, or equivalent
- Cleaning solvent, Part Number 5298527, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : Brush away all loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

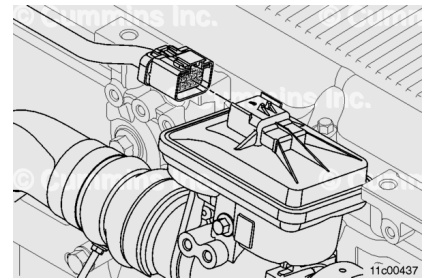
Note : Some chassis may require the removal of the exhaust gas recirculation (EGR) mass measurement flow assembly and exhaust transfer elbow connection to better gain access to the EGR valve.

- Disconnect the batteries. See equipment manufacturer service information.
- If required, remove the EGR mass measurement flow assembly. Refer to Procedure 011-066 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-066.html>)
- If required, remove the exhaust transfer elbow connection. Refer to Procedure 011-068 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-068.html>)
- Disconnect the wiring harness from the EGR valve mounting bracket attachment points around the EGR valve.

Remove

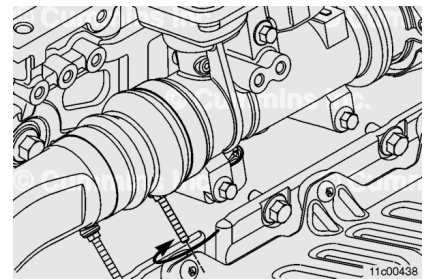
Disconnect the EGR valve electrical connector.

Disconnect the wire harness from the EGR valve mounting location.



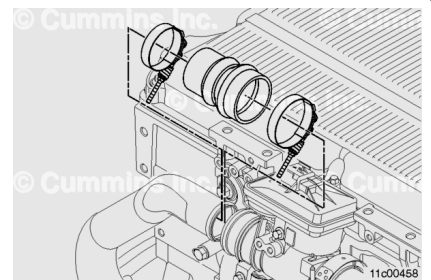
Loosen the hose clamp on the EGR valve.

Note : For ease of removal, run a hose pick under the EGR system hose on the inlet side of the EGR valve.



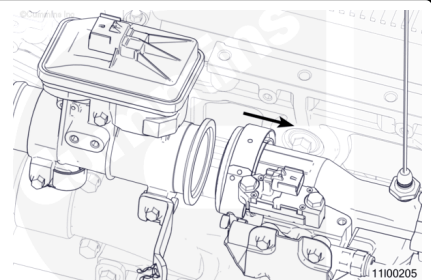
Remove the EGR system hose from the rear EGR connection tube. Refer to Procedure 011-034 in Section 11.

(/qs3/pubsys2/xml/en/procedures/377/377-011-034.html)

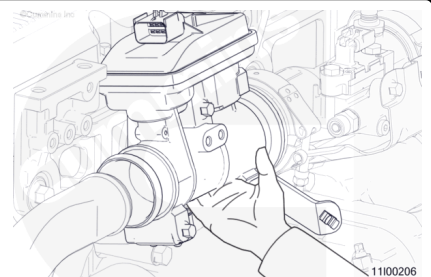


If the mass measurement flow assembly has **not** been removed, loosen the V-band clamp between the EGR valve and the mass measurement flow assembly.

Slide the V-band over the intake side of the mass measurement flow assembly.



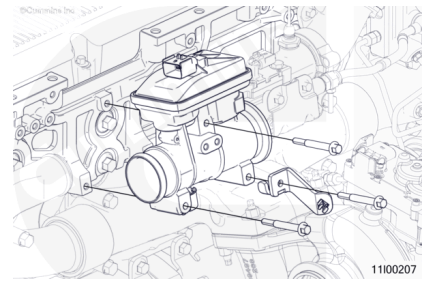
Make sure to support the EGR valve by hand before removing the mounting capscrews. Otherwise, it can fall off the engine, causing damage to the EGR valve.



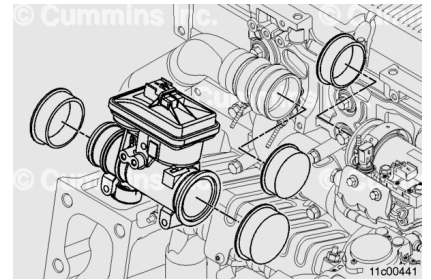
Loosen and remove the EGR valve mounting capscrews.

Remove the EGR valve from the EGR valve support bracket.

Remove and discard the o-ring from the inlet side of the EGR mass measurement flow assembly.



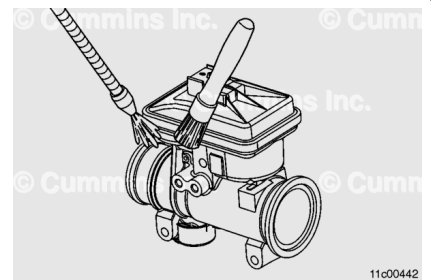
Use protective caps from the Air Handling Clean Care Kit, Part Number 4918588, or equivalent, to cover open points on the EGR valve and plumbing.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR valve mounting and sealing surfaces with safety solvent to remove any deposits or debris.

Clean the accumulated soot from the EGR valve inlet port/outlet port using a wire brush. Soak with cleaning solvent, Part Number 5298527, or equivalent, and clean again with a wire brush.

Rinse the valve body with clean water. Dry with compressed air.

Make sure the loose soot is removed from the EGR valve housing. This can be done by vacuuming the soot out of the EGR valve inlet/outlet port.

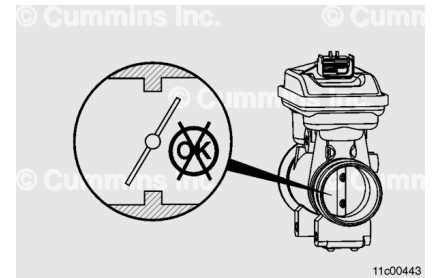
Note : Valves with heavy carbon buildup may need to be cleaned with a scraper prior to using solvent and a wire brush. If this is necessary, make sure **not** to pry against the butterfly valve during the cleaning process.

Inspect the mounting surfaces for cracks, fretting, or other damage.

Inspect the EGR butterfly valve for a stuck open condition. If the EGR valve is stuck open, replace the EGR valve. Do **not** try to pry the butterfly open or closed.

Inspect the inside of the valve for cracks, broken butterfly valve, or other damage.

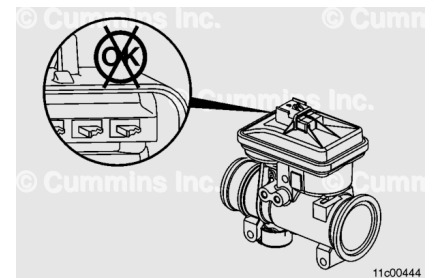
Replace the EGR valve if any damage is found.



Inspect the EGR valve actuator for cracks or other damage. Replace if damage is found.

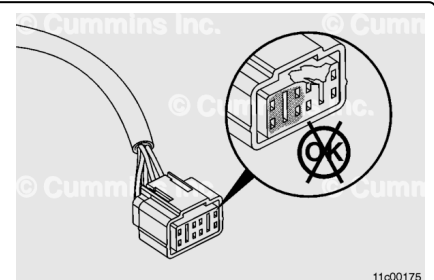
Inspect the EGR valve actuator connector for corrosion, worn pins, or other damage.

Replace the EGR valve assembly if corrosion, worn pins, or other damage is found.

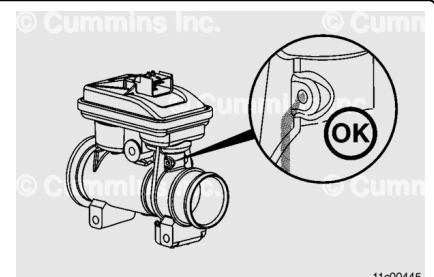


Inspect the engine wiring harness EGR valve actuator connector for corrosion, worn receptacles, or other damage.

Replace the engine wiring harness EGR valve connector if corrosion, worn receptacles, or other damage is found.

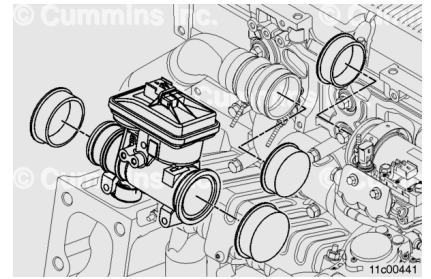


Soot leakage at the EGR valve weep holes is normal and is to be expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep holes.



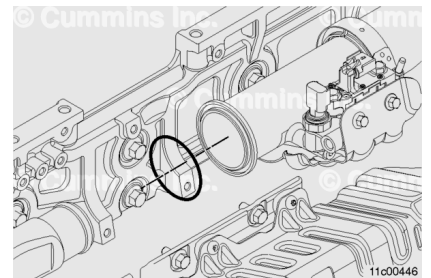
Install

Remove the protective caps from the Air Handling Clean Care Kit, Part Number 4918588, or equivalent, used to cover open points on the EGR valve and plumbing.



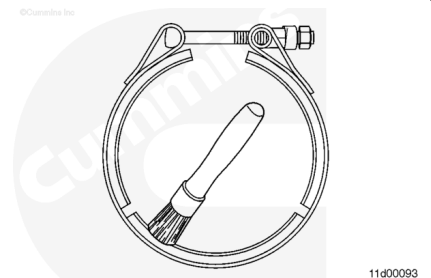
Push the o-ring over the flow assembly pilot by hand.

The o-ring **must** remain square during installation. Do **not** allow the o-ring to roll or twist.



Before installing the EGR valve, coat the clamping surface of the V-band with high temperature anti-seize compound.

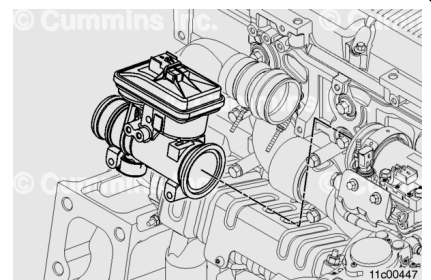
Slide the V-band over the intake side of the mass measurement flow assembly if it was **not** removed from the engine.



Install the EGR system hose. Refer to Procedure 011-034 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-034.html>)

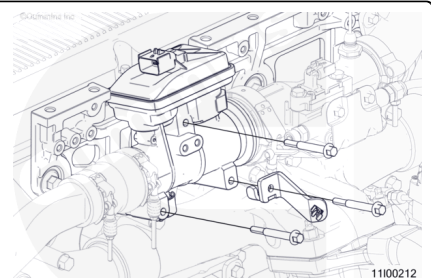
Apply lubricant to the hose end of the EGR valve. Slide the hose clamp over the EGR hose.

Slide the EGR valve into the EGR system hose. Make sure that the hose slides past the hose bead on the EGR valve.



Align the EGR valve mounting holes with their respective holes in the EGR valve bracket.

Align the wire attachment bracket on the hole near the EGR mass measurement assembly.

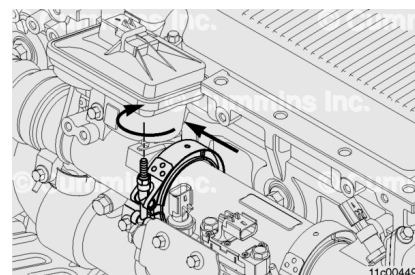


Hand-tighten the three EGR valve mounting capscrews to hold the EGR valve in place.

Slide the V-band in place over the EGR valve outlet and EGR mass measurement flow assembly inlet.

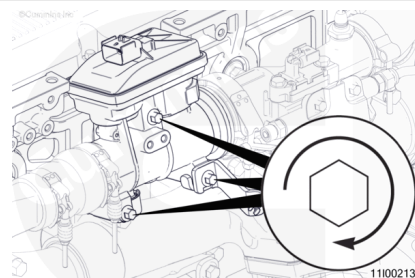
Tighten the V-band clamp.

Torque Value: 12 n•m [106 in-lb]

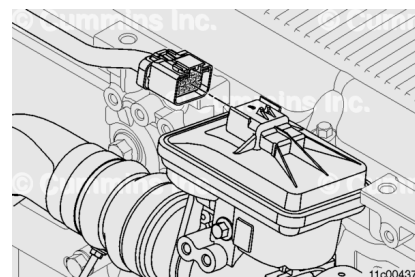


Tighten the mounting capscrews.

Torque Value: 20 n•m [177 in-lb]

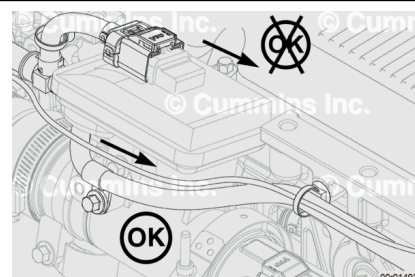


Connect the EGR valve connector.



Route the EGR valve wiring harness and the EGR outlet temperature sensor wiring harness from the EGR valve mounting bracket attachment points around the EGR valve.

Do **not** route the EGR outlet temperature sensor wire harness on top of the EGR valve.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the wiring harness from the EGR valve mounting bracket attachment points around the EGR valve.
- If removed, install the exhaust transfer elbow connection. Refer to Procedure 011-068 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-068.html>)
- If removed, install the EGR mass measurement flow assembly. Refer to Procedure 011-066 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-066.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.